Counting and Sampling - Examples

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Mini-batch Sampling A training dataset consists of 100 examples, of which 60 belong to class A and 40 belong to class B. A mini-batch of size 5 is constructed by sampling examples independently and uniformly at random from the dataset *with replacement*. Thus, the same training example may be selected more than once.

(a) What is the probability that all 5 examples in the mini-batch are distinct?

(b) What is the probability that the mini-batch contains at least one example from each of the two classes?

Solution

(a) Since sampling is with replacement, there are 100 possible choices on each draw.

The first selected example can be arbitrary. For all examples to be distinct, the second example must be different from the first, the third must be different from the first two, and so on. Therefore,

$$\Pr(\text{all 5 examples are distinct}) = 1 \cdot \frac{99}{100}\frac{98}{100}\frac{97}{100}\frac{96}{100}.$$

Equivalently,

$$\boxed{\Pr(\text{all 5 examples are distinct}) = \frac{100 \cdot 99 \cdot 98 \cdot 97 \cdot 96}{100^5}.}$$

(b) On each draw, the probabilities of selecting an example from classes A and B are

$$\Pr(A) = \frac{60}{100} = 0.6, \qquad \Pr(B) = \frac{40}{100} = 0.4.$$

The mini-batch fails to contain both classes precisely when either all 5 examples are from class A or all 5 examples are from class B.

Since the draws are independent,

$$\Pr(\text{all 5 are from A}) = (0.6)^5$$

and

$$\Pr(\text{all 5 are from B}) = (0.4)^5.$$

These two events are disjoint. Hence,

$$\Pr(\text{only one class is represented}) = (0.6)^5 + (0.4)^5.$$

Taking the complement,

$$\boxed{\Pr(\text{both classes are represented}) = 1 - (0.6)^5 - (0.4)^5.}$$

Dataset Quality Check

A machine-learning dataset contains 1000 labeled images. It is known that 50 of these images have incorrect labels. As part of a dataset quality check, a data scientist randomly selects 20 distinct images from the dataset for manual inspection.

Let X denote the number of incorrectly labeled images found among the 20 inspected images.

(a) What is the probability that the quality check finds no incorrectly labeled images?

(b) What is the probability that the quality check detects at least one incorrectly labeled image?

(c) What is the probability that exactly two incorrectly labeled images are found?

Solution

There are

$$\binom{1000}{20}$$

equally likely ways of selecting the 20 images.

(a) There are 950 correctly labeled images. For the quality check to find no labeling errors, all 20 inspected images must be chosen from these 950 images. Hence,

$$\boxed{\Pr(X = 0) = \frac{\binom{950}{20}}{\binom{1000}{20}}.}$$

(b) It is convenient to use the complement of the event in part (a):

$$\Pr(X \geq 1) = 1 - \Pr(X = 0).$$

Therefore,

$$\boxed{\Pr(X \geq 1) = 1 - \frac{\binom{950}{20}}{\binom{1000}{20}}.}$$

(c) For exactly two incorrectly labeled images to be found, we must choose 2 images from the 50 incorrectly labeled images and the remaining 18 images from the 950 correctly labeled images. Thus, the number of such samples is

$$\binom{50}{2}\binom{950}{18}.$$

Hence,

$$\boxed{\Pr(X = 2) = \frac{\binom{50}{2}\binom{950}{18}}{\binom{1000}{20}}.}$$

More generally, the probability of finding exactly r incorrectly labeled images in the sample is

$$\boxed{\Pr(X = r) = \frac{\binom{50}{r}\binom{950}{20-r}}{\binom{1000}{20}}.}$$

Classification Errors

A classifier makes an error on any given test example with probability 0.2. Suppose the classifier is evaluated on 10 independently selected test examples, and assume that its errors on the different examples are independent.

Let X denote the total number of classification errors among the 10 test examples.

(a) What is the probability that the classifier makes exactly 3 errors?

(b) Given that the classifier makes exactly 3 errors, what is the probability that at least two of the errors occur on consecutive test examples?

Solution

(a) Consider any particular choice of 3 test examples on which the classifier makes errors. The probability that the classifier makes errors on these 3 examples and is correct on the remaining 7 is

$$(0.2)^3(0.8)^7.$$

There are

$$\binom{10}{3}$$

ways to choose the 3 examples on which the classifier makes errors. Therefore,

$$\boxed{\Pr(X = 3) = \binom{10}{3}(0.2)^3(0.8)^7.}$$

(b) Define the events

$$A = \{\text{at least two errors occur consecutively}\}$$

and

$$B = \{X = 3\}.$$

We want to compute

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}.$$

We already know from part (a) that

$$\Pr(B) = \binom{10}{3}(0.2)^3(0.8)^7.$$

To compute $\Pr(A \cap B)$, it is convenient to first count the complement: the number of ways in which exactly 3 errors can occur with no two errors consecutive.

If there are exactly 3 errors, there are 7 correct classifications. Place these 7 correct classifications first:

$$C \quad C \quad C \quad C \quad C \quad C \quad C.$$

They create 8 possible gaps:

$$_C_C_C_C_C_C_C_.$$

In order that no two errors be consecutive, at most one error can be placed in each gap. Thus, the number of ways to place the 3 errors with no two consecutive is

$$\binom{8}{3}.$$

Since there are $\binom{10}{3}$ ways to choose the 3 error positions in total, the number of error patterns having at least two consecutive errors is

$$\binom{10}{3} - \binom{8}{3}.$$

Each such pattern has probability $(0.2)^3(0.8)^7$. Hence,

$$\Pr(A \cap B) = \left(\binom{10}{3} - \binom{8}{3} \right) (0.2)^3(0.8)^7.$$

Therefore,

$$\begin{aligned} \Pr(A \mid B) &= \frac{\left(\binom{10}{3} - \binom{8}{3} \right) (0.2)^3(0.8)^7}{\binom{10}{3}(0.2)^3(0.8)^7} \\ &= 1 - \frac{\binom{8}{3}}{\binom{10}{3}} \\ &= 1 - \frac{56}{120} \\ &= \boxed{\frac{8}{15}}. \end{aligned}$$